Vishwakarma Institute of Technology Issue 01 : Rev No. 00 : Dt. 01/08/22

Title : Question Paper

FF No. 868

Reg No.							

Bansilal Ramnath Agarwal Charitable Trust's
VISHWAKARMA INSTITUTE OF TECHNOLOGY, PUNE – 411037.
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Examination: ESE

Year: TY **Branch**: Computer

Subject: Design and Analysis of Algorithms **Subject Code:** CS 3205

Max. Marks:60 **Total Pages of Question Paper:** 3

Day & Date: wed. 08/5/24 **Time:** 02:30 pm to 4:30pm

Instructions to Candidate

1. Figures to the right indicate full marks.
2. Try to avoid using informal arguments. They won't get any credits. Try to write formal, concrete, and concise answers.
3. Use simple English language along with simple programming constructs while describing the algorithm. If the examiner cannot understand the programming constructs or the syntax you used, then naturally you will not get credit. Your goal is to convey your algorithm to the examiner in simplest possible language (of course without compromising concreteness, correctness)
4. While describing algorithms you must use only simple data structures like arrays, linked list, trees, etc. Any use of any advanced data structures like sets, hashmap, etc carry zero weightage.

Q. N.	CO No	BT* No		Max marks
Q. 1.				
Q.1. (a)	3	2, 3	For the following two recurrences give best possible asymptotic upper bound on T(n) as a function of n. Express your answer using Big Oh asymptotic notation. (1 mark for correct final answer and 2 marks for the correct approach) (1) T(n) = 3 T (n - 1) + 1 (2) T(n) = 9 T (n/3) + n^2.	6 marks 6
Q.1. (b)			Attempt Any **ONE** of the following.	4 marks
Q.1. (b)(i)	3	2, 3	Formally define Big Theta asymptotic notation and prove that H(n) = Theta (log n). Where H(n) = 1 + 1/2 + 1/3 + 1/4 + ... + 1/n. (that is H(n) is sum of reciprocals of first n natural numbers)	2
Q.1. (b)(ii)	1, 3	1,3	Let G be a simple connected weighted graph. Prove that there exists a minimum weight spanning tree of G which contains a minimum weight edge of G. (Note: a concrete proof from scratch is expected)	
Q. 2.				
Q. 2. (a)	1, 2, 3	1, 3	A. Let f(n) denotes number of binary trees on n nodes (nodes are not labeled). Give a recurrence relation for f(n). (2 marks) B. A full binary tree is a binary tree in which every node has either zero or two children. Let B(n) denote the number of full binary trees on n nodes (nodes are not labeled). (1) Give a recurrence relation for B(n) (2) Prove that if n is an even number, then B(n) = 0 (3) Using the recurrence above or otherwise determine B(n) for all n = 3, 5, 7, 9.	8 marks 2
Q. 2. (b)	4, 6	1	Give prover-verifier based definition of complexity class NP and give two concrete examples of problems characterizing NP class.	2 marks 2
Q.3.				
Q.3	2, 3	2	Consider a text over alphabet {a,b,c,d,e,f,g} relative frequencies of the	6 marks 6

(a)			characters are 5, 7, 8, 16, 18, 22, 24 respectively. Construct Huffman tree. Find Huffman encoding for each of the seven characters. Suppose original text contains total 1,00,000 characters what will be size of the text in bits after encoding. (5 marks for Huffman tree and correct codes, 1 mark for correct text size after encoding)	8 marks
Q. 3. (b)			Attempt any **ONE** of the following	8
Q. 3. (b) (i)	1,2,3,6	3, 4	Given integers n and k, and numbers p_1, ..., p_n ∈ [0, 1] (that is 0 <= p_i <= 1 for every i). You want to determine the probability of obtaining exactly k heads when n biased coins are tossed independently at random, where p_i is the probability that the ith coin comes up heads. Give an O(n^2) time algorithm for this task with proper Pseudo code. Give an argument justifying correctness of the algorithm and the time complexity. Assume you can multiply and add two numbers in [0, 1] in O(1) time.	
Q. 3. (b) (ii)	1, 2, 3	3, 4	Let M_1, M_2, ..., M_k be the matrices such that dimension of matrix M_i is **n_i** x **n_{i+1}** for 1 <= i <= k. There are several different ways to compute the matrix product M_1M_2...M_k. Give an efficient algorithm to find out a best way (which uses minimum number of integer multiplications) to compute the product M_1M_2...M_k. Your algorithm shall output a root node of the binary tree T such that every internal node of the tree is a multiplication gate and leaf nodes are the matrices M_1, M_2, ..., M_k such that the evaluation of the tree corresponds to a best way to compute the product M_1M_2...M_k	
Q.4.	1, 2,3 5,6	4, 6	A vertex cover of a graph G(V,E) is a set of vertices S such that every edge of G has at least one end point in S. A tree is an undirected connected graph which doesn't have any cycles. Given a tree T on n vertices as input, give an efficient algorithm for finding size of a smallest sized vertex cover for T. Give formal pseudocode and analyze the time complexity of the algorithm. Full credit will be given only for the correct algorithm whose time complexity is asymptotically equal to the time complexity of the best algorithm for the problem.	6 marks
Q.5.	2, 3, 5, 6	3, 4	1. Let A[1-n] be array of size n containing distinct integers. Let S be array of size n such that S[i] = length of longest increasing subsequence in A which ends at index i for 1<=i<=n. Give a concrete recurrence relation for S[i]. Describe the base case of the recurrence. Express the length of longest increasing sequence of A in terms of entries of S. 2. Let A[0-n] be array of size n+1 such that A[1] to A[n] contain n distinct integers and A[0] is some integer smaller than A[1], A[2], ..., A[n]. For 1<= i<=n and 0<= j <= n let f(i,j) be the length of a longest increasing subsequence in A which starts at index i or onwards and has all the elements greater than or equal to A[j]. Give a concrete recurrence for f(i,j). Describe the base case of the recurrence. If we use the recurrence to construct a dynamic programming solution for the longest increasing subsequence problem, what will be time and the space complexity of the resulting algorithm.	8 marks
Q.6.				
Q. 6. (a)	1, 2, 3, 6	4, 6	You are given two **sorted** arrays A and B each containing n distinct integers and a natural number r <= 2n as input. Assume that there is no common element in A and B. Let C be set of all elements either in A or B (that is C is union of A and B). For element p in array C rank(p) in C is a total number of elements in C which are less than or equal to p. (E.g. if C = {2, 7, 1, 9} then rank(7) in C is 3 as elements 1, 2, 7 in C are less than or equal to 7). Give an efficient algorithm to find an element p in C with rank(p) = r. Give proper Pseudo-code for the algorithm. Analyze the time complexity of the algorithm. For full credit time complexity of your algorithm must be asymptotically equal to the time complexity of the best algorithm for the problem.	7 marks 7
Q. 6.	1, 2, 3	3, 4	Let T be a binary tree with n nodes such that each node of the tree stores a	5 marks

			number. The weight of a path is the sum of the numbers corresponding to the nodes on the path. Give an efficient algorithm to find weight of a best root to leaf path in T (a best path is the one with smallest weight). Give a best possible upper-bound on time complexity of your algorithm.	

Statements:

1. To formulate computational problems in abstract and mathematically precise manner
2. To design efficient algorithms for computational problems using appropriate algorithmic paradigm
3. To analyze asymptotic complexity of the algorithm for a complex computational problem using suitable mathematical techniques.
4. To establish NP-completeness of some decision problems, grasp the significance of the notion of NP-completeness and its relationship with intractability of the decision problems.
5. To understand significance of randomness, approximability in computation and design randomized algorithms for simple computational problems and design efficient approximation algorithms for standard NP-optimization problems.
6. To incorporate appropriate data structures, algorithmic paradigms to craft innovative scientific solutions for complex computing problems.

ms Taxonomy (BT) Level No:

nembering; 2. Understanding; 3. Applying; 4. Analyzing; 5. Evaluating; 6. Creating